

MUEHLEBERG, SwitzerlandWMO: **066360**Lat: **46.9733N**Lon: **7.2781E**Elev: **1585**StdP: **13.87**Time Zone: **1.00 (EUC)**Period: **08-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 16.4	(c) 20.9	(d) 10.7	(e) 10.0	(f) 18.8	(g) 16.5	(h) 13.4	(i) 23.6	(j) 16.0	(k) 44.6	(l) 13.3	(m) 42.1	(n) 4.5	(o) 90	(p) 0.462

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 20.1	(c) 86.8	(d) 68.9	(e) 83.3	(f) 67.7	(g) 80.0	(h) 66.1	(i) 70.5	(j) 82.5	(k) 69.0	(l) 80.3	(m) 67.5	(n) 77.4	(o) 4.2	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.8	104.6	74.8	65.3	99.3	72.7	64.0	94.6	70.9	35.4	82.8	34.0	80.9	32.8	77.7	79.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.1	10.4	8.9	12.7	92.1	4.8	2.8	9.3	94.1	6.4	95.7	3.7	97.3	0.2	99.3
			12.2	74.4	4.9	2.1	8.7	75.9	5.9	77.1	3.1	78.3	-0.4	79.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.4	33.7	34.6	42.7	50.5	56.2	63.9	67.5	66.5	59.4	51.2	42.2	35.2		
	DBStd	13.49	6.61	7.30	6.18	6.18	6.31	5.97	5.38	5.41	5.43	6.29	6.03	6.17		
	HDD50	2022	505	430	234	70	16	0	0	0	2	65	240	460		
	HDD65	5617	969	850	692	436	280	91	37	48	180	426	683	925		
	CDD50	2167	0	0	7	84	208	418	541	513	286	103	7	0		
	CDD65	287	0	0	0	0	6	59	113	96	13	0	0	0		
	CDH74	2561	0	0	0	12	106	570	1018	761	93	1	0	0		
	CDH80	707	0	0	0	0	14	152	319	216	6	0	0	0		
(14) Wind	WSAvg	2.8	2.9	3.0	3.4	3.2	3.1	2.9	2.7	2.3	2.5	2.2	2.5	2.7		
(15) Precipitation	PrecAvg	46.0	2.5	2.5	2.8	3.5	5.0	5.1	5.3	4.8	4.0	3.9	3.3	3.2		
	PrecMax	57.9	6.1	5.8	10.3	7.8	8.9	10.6	11.5	8.5	6.9	6.9	7.1	6.4		
	PrecMin	35.6	0.7	0.5	1.1	0.7	1.9	2.1	1.9	0.7	2.0	0.6	0.2	0.2		
	PrecStd	6.3	1.5	1.5	1.9	1.7	2.0	2.0	2.0	1.9	1.4	1.5	1.9	1.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.6	57.7	65.6	75.8	81.9	88.9	91.7	89.6	80.7	70.8	61.3	52.7		
		MCWB	47.9	47.9	51.1	59.2	63.7	70.7	70.2	69.3	65.2	59.1	53.9	48.4		
	2%	DB	49.0	52.1	62.5	71.1	76.8	84.3	87.3	85.5	77.0	67.3	57.6	50.1		
		MCWB	45.0	45.0	50.1	55.5	61.9	68.5	68.1	69.2	64.2	58.7	52.0	46.2		
	5%	DB	45.3	48.2	58.5	68.0	73.0	80.5	83.5	82.0	73.0	63.5	53.7	46.9		
		MCWB	42.3	43.3	48.3	54.8	59.5	66.8	67.3	68.0	62.5	57.4	50.1	44.0		
	10%	DB	42.4	44.2	54.2	63.7	69.0	76.7	79.8	78.2	69.5	60.4	50.6	43.9		
		MCWB	39.9	40.2	46.4	52.8	58.1	64.3	65.7	66.3	61.1	56.0	47.8	41.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.1	48.6	52.8	59.3	66.5	74.0	72.6	71.9	67.7	62.1	54.8	49.1		
		MCDB	52.0	55.2	62.4	72.6	78.1	85.8	86.9	83.4	77.3	66.7	59.6	51.8		
	2%	WB	45.2	45.5	50.9	56.7	63.4	70.0	70.1	70.2	65.6	59.9	52.6	46.8		
		MCDB	47.6	51.3	60.1	69.1	73.4	81.5	82.7	82.1	73.4	64.4	56.4	49.2		
	5%	WB	42.5	43.1	49.1	55.1	60.9	67.9	68.6	68.9	63.8	58.3	50.5	44.4		
		MCDB	45.2	47.6	57.3	65.9	69.6	78.3	79.8	79.7	70.6	62.1	53.5	46.6		
	10%	WB	40.4	40.5	47.1	53.4	58.9	65.8	67.0	67.3	62.2	56.5	48.3	42.0		
		MCDB	42.3	44.0	53.4	62.9	67.1	74.5	77.0	76.4	68.0	59.7	50.1	43.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.8	12.1	16.8	19.2	18.2	18.9	20.1	19.5	17.4	13.9	10.1	8.7		
		MCDBR	12.9	19.6	25.4	27.3	25.7	25.8	27.0	25.4	22.4	18.7	15.2	12.1		
		MCWBR	9.7	12.9	14.1	13.2	11.7	11.8	10.3	10.5	10.9	10.9	10.1	9.0		
	5% WB	MCDBR	12.3	18.5	23.1	25.7	23.2	24.4	24.6	23.7	20.0	16.5	14.4	11.3		
		MCWBR	9.6	12.5	13.1	12.7	11.5	11.6	10.3	10.4	10.3	10.4	9.9	8.9		
(40) Clear-Sky Solar Irradiance	taub		0.294	0.321	0.359	0.398	0.410	0.411	0.411	0.401	0.379	0.355	0.320	0.293		
	taud		2.432	2.361	2.309	2.231	2.247	2.288	2.284	2.322	2.365	2.416	2.456	2.442		
	Ebn at Noon		254	268	272	271	272	271	270	267	263	252	241	240		
	Edh at Noon		22	28	35	41	42	41	40	37	33	27	21	19		
(44) All-Sky Solar Radiation	RadAvg		392	660	1033	1382	1590	1884	1829	1591	1229	784	429	318		
	RadStd		48	100	154	202	147	176	160	147	112	77	55	55		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (112 neighbors)	N/A	N/A	N/A	+1.35	+0.72	+0.54	N/A	N/A	+104	+32

Nomenclature: See separate page